

Sentiment Analysis of Customer Product Reviews Using Transformer-Based Language Models

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Abstract—Online shopping platforms receive millions of customer reviews every day, and manually reading them to understand customer opinion is not practical. Sentiment analysis, a core task of Natural Language Processing (NLP), automatically identifies whether a piece of text expresses a positive, negative or neutral opinion. Traditional approaches based on Bag-of-Words features and classical machine learning classifiers fail to capture context, negation and sarcasm present in real reviews. This paper proposes a sentiment classification system for e-commerce product reviews that fine-tunes a pre-trained transformer model, BERT, and compares it with classical baselines (TF-IDF with Logistic Regression and Naive Bayes) and a recurrent baseline (Bi-LSTM with GloVe embeddings). The paper describes the dataset, the text preprocessing pipeline, the model architecture, the training strategy and the evaluation metrics (accuracy, precision, recall and F1-score). The proposed system is intended to help sellers and platforms summarise customer feedback quickly and take data-driven product decisions.

Keywords—Natural Language Processing, Sentiment Analysis, BERT, Transformers, Deep Learning, Product Reviews, Text Classification

I. INTRODUCTION

The rapid growth of e-commerce has changed the way people buy products. Before purchasing, most customers read the reviews written by earlier buyers, and sellers depend on the same reviews to understand what customers like or dislike about a product. Popular products can collect thousands of reviews, which makes manual analysis slow, costly and inconsistent. Sentiment analysis, also known as opinion mining, is the branch of Natural Language Processing (NLP) that automatically extracts the polarity of an opinion from text [1].

Early work treated sentiment analysis as a standard text classification problem and applied Naive Bayes, Maximum Entropy and Support Vector Machine classifiers on word-count features [2]. These methods are fast and simple but ignore word order, so sentences such as “not bad at all” or “I expected much better” are often misclassified. Deep learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks improved results by learning features from word embeddings [4]–[7]. More recently, transformer-based language models pre-trained on large text corpora, such as BERT [9], have become the standard approach for most NLP tasks because they model the full context of every word in both directions.

The main objectives of this work are:

- To build a clean, reproducible preprocessing pipeline for noisy product review text.

- To fine-tune a pre-trained BERT model for three-class (positive, neutral, negative) review sentiment classification.
- To compare the transformer model with classical machine learning and Bi-LSTM baselines using standard evaluation metrics.
- To design a simple interface through which a user can enter a review and obtain its predicted sentiment with a confidence score.

The rest of the paper is organised as follows. Section II reviews related work, Section III describes the proposed methodology and Section IV explains the system architecture and evaluation plan.

II. LITERATURE SURVEY

Pang et al. [2] were among the first to apply machine learning to sentiment classification of movie reviews and showed that unigram features with SVM performed better than hand-crafted rules. Maas et al. [3] released the widely used IMDb Large Movie Review dataset and learned word vectors that capture sentiment information. Word embedding techniques such as Word2Vec [4] and GloVe [5] represent words as dense vectors so that semantically similar words lie close to each other, which became the input layer for most neural models.

Kim [6] showed that a simple CNN with one convolution layer over pre-trained word vectors achieves strong results on several sentence classification benchmarks. LSTM networks [7] handle long-range dependencies in text and have been widely used for review sentiment analysis. Socher et al. [13] introduced the Stanford Sentiment Treebank and demonstrated that modelling sentence structure helps to capture negation.

The transformer architecture proposed by Vaswani et al. [8] replaced recurrence with self-attention and allowed efficient training on very large corpora. Devlin et al. [9] introduced BERT, which is pre-trained using masked language modelling and can be fine-tuned for downstream tasks by adding a single output layer. RoBERTa [10] improved BERT through better pre-training choices, while DistilBERT [11] reduced model size for faster inference. McAuley and Leskovec [12] collected a large Amazon product review corpus that is commonly used for review-level research. Based on this survey, fine-tuned transformer models are selected as the core of the proposed system, with classical and recurrent models as baselines.

III. PROPOSED METHODOLOGY

A. Dataset

The system uses publicly available Amazon product reviews [12]. Each review contains the review text and a star rating from 1 to 5. Ratings of 1–2 are labelled negative, 3 is labelled neutral and 4–5 are labelled positive. A balanced subset is sampled so that each class is equally represented, and the data is split into training (80%), validation (10%) and test (10%) sets.

B. Text Preprocessing

Raw review text contains HTML tags, URLs, emojis, repeated characters and spelling mistakes. The preprocessing steps are: (i) removal of HTML tags and URLs, (ii) lower-casing (for baseline models), (iii) expansion of contractions such as “don’t” to “do not” so that negation is preserved, (iv) tokenisation, and (v) stop-word removal and lemmatisation for the classical baselines only. For BERT, the WordPiece tokenizer is applied directly and stop words are kept, because the model relies on the full sentence context.

C. Baseline Models

Two classical baselines are trained on TF-IDF features of unigrams and bigrams: Multinomial Naive Bayes and Logistic Regression. A deep learning baseline uses a two-layer Bidirectional LSTM with 300-dimensional GloVe [5] embeddings followed by a dense softmax layer.

D. Transformer Model

The main model is the pre-trained bert-base-uncased model [9], which has 12 encoder layers, 768 hidden units and 12 attention heads. Each review is converted to the form [CLS] tokens [SEP], truncated or padded to 256 tokens. The final hidden state of the [CLS] token is passed through a dropout layer and a linear layer with softmax activation to produce probabilities for the three classes. The complete network is fine-tuned end-to-end using the AdamW optimiser with a small learning rate, linear warm-up and cross-entropy loss. Early stopping on the validation F1-score is used to prevent over-fitting.

E. Evaluation Metrics

All models are evaluated on the same held-out test set using accuracy, precision, recall and macro-averaged F1-score. A confusion matrix is used to study the errors between neighbouring classes, especially between neutral and the two polar classes, which is known to be the most difficult case.

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